

Transaction ID	Salesperson	Region	Product	Category	Date	Revenue (\$)	Cost (\$)
1001	Alice	North	Laptop	Electronics	2024-01-12	1500	1000
1002	Bob	South	Phone	Electronics	2024-02-18	800	500
1003	Charlie	East	Laptop	Electronics	2024-02-25	1600	1100
1004	Alice	North	Chair	Furniture	2024-03-05	300	150
1005	Bob	West	Desk	Furniture	2024-03-18	500	300
1006	Charlie	East	Phone	Electronics	2024-04-10	900	600
1007	Alice	North	Desk	Furniture	2024-04-22	550	350
1008	Bob	South	Laptop	Electronics	2024-05-14	1700	1200
1009	Charlie	East	Chair	Furniture	2024-05-28	350	200
1010	Alice	West	Phone	Electronics	2024-06-03	750	450

1. Creating a Pivot Table

Create a pivot table to summarize the total revenue for each region. What steps would you follow?

Ans : Please follow below steps :

1. Select the dataset and click on Insert -> Pivot table option
2. Once the table is inserted, in the row select the region option
3. In the value select the Revenue summarized by Sum option.

Region	SUM of Revenue (\$)
East	2850
North	2350
South	2500
West	1250
Grand Total	8950

2. Filtering Data

Using a pivot table, how can you filter the dataset to show only sales for Electronics?

Ans : Please follow below steps :

1. Select the dataset and click on Insert -> Pivot table option
2. Once the table is inserted, in the row select the Category option
3. In the value select the Revenue summarized by Sum option.
4. In the filter option select the Category as Electronics.

Category	SUM of Revenue (\$)
Electronics	7250
Grand Total	7250

3. Sorting Revenue in Descending Order

How can you sort the pivot table so that regions with the highest revenue appear at the top?

Ans : Please follow below steps :

1. Select the dataset and click on Insert -> Pivot table option
2. Once the table is inserted, in the row select the region option
3. In the value select the Revenue summarized by Sum option.
4. In the row field select the order as descending and sort by as->Sum of Revenue

Region	SUM of Revenue (\$)
East	2850
South	2500
North	2350
West	1250
Grand Total	8950

4. Grouping Data by Month

How would you group the transactions by month to analyse sales trends over time?

Ans : Please follow below steps :

1. Select the dataset and click on Insert -> Pivot table option
2. Once the table is inserted, in the row select the date option
3. In the value select the Revenue summarized by Sum option.
4. Right Click on any date field and select the Create pivot date group -> Months

Date - Month	SUM of Revenue (\$)
Jan	1500
Feb	2400
Mar	800
Apr	1450
May	2050
Jun	750
Grand Total	8950

5. Adding a Calculated Field

Create a calculated field in the pivot table to determine Profit (Revenue - Cost) for each salesperson.

Ans : Please follow below steps :

1. Select the dataset and click on Insert -> Pivot table option
2. Once the table is inserted, in the row select the Salesperson option
3. In the value field select Revenue and Cost fields summarized by SUM
4. In the value field select Calculate field and mention formula as 'Revenue (\$)' – Cost (\$)'

The screenshot shows a Google Sheets document titled "Copy of Sales Transaction Data". The main data table is as follows:

Region	SUM of Revenue	Category	SUM of Revenue (\$)
East	2850	Electronics	7250
South	2500		
North	2350		
West	1250		
Grand Total	8950		7250

Below this, a second Pivot Table is shown with the following data:

Salesperson	SUM of Revenue	SUM of Cost (\$)	Calculated Field
Alice	3100	1950	1150
Bob	3000	2000	1000
Charlie	2850	1900	950
Grand Total	8950	5850	3100

The Pivot Table Editor on the right shows the following configuration:

- Columns:** Transaction ID, Salesperson, Region, Product, Category, Date, Revenue (\$), Cost (\$)
- Values:**
 - Revenue (\$):** Summarize by SUM, Show as Default
 - Cost (\$):** Summarize by SUM, Show as Default
 - Calculated Field 1:** Formula: =Revenue (\$)-Cost (\$), Summarize by SUM, Show as Default

6. Using Slicers for Interactive Filtering

How would you use a slicer to filter data by a Salesperson so that you can quickly compare performance?

Ans : Please follow below steps :

1. Select the dataset and click on Insert -> Pivot table option
2. Once the table is inserted, in the row select the Salesperson option
3. In the value field select Revenue and Cost fields summarized by SUM
4. In the value field select Calculate field and mention formula as = 'Revenue (\$)' - Cost (\$)'
5. Select the created pivot table and navigate to Data-> Add a Slicer and select Salesperson in column
6. Now when selecting a salesperson , the performance would be reflected.

The screenshot shows a Google Sheets interface with a Pivot table and its editor. The Pivot table is located in the range F22:G24 and has the following data:

Salesperson	SUM of Revenue (\$)	SUM of Cost (\$)	Calculated Field
Alice	3100	1950	1150

The Pivot table editor is open on the right side of the screen. It shows the following settings:

- Rows:** Salesperson
- Columns:** (Empty)
- Values:** Revenue (\$), Cost (\$)
- Calculate field:** Revenue (\$) - Cost (\$)

The editor also shows a list of available fields on the right, including Transaction ID, Salesperson, Region, Product, Category, Date, Revenue (\$), and Cost (\$).

7. Identifying the Best-Selling Product

Using the pivot table, determine which Product category generated the highest total revenue.

Ans : Please follow below steps :

1. Select the dataset and click on Insert -> Pivot table option
2. Once the table is inserted, in the row select the Product option
3. In the value field select Revenue and Cost fields summarized by SUM
4. In the row select order as descending and sort by as SUM of revenue option
5. The top row will determine the product category having highest total revenue.

Product	SUM of Revenue
Laptop	4800
Phone	2450
Desk	1050
Chair	650

8. Applying Conditional Formatting

Apply conditional formatting to highlight transactions where revenue is greater than \$1,500.

Ans : 1. From the provided dataset select the Revenue (\$) column and select

Format->Conditional Formatting

2. Select the column range on Revenue and apply format rules as value>1500 then highlight it with a selected colour(Green)

The screenshot shows a Google Sheets interface with a dataset of sales transactions. The dataset is organized into columns: Transaction ID, Salesperson, Region, Product, Category, Date, Revenue (\$), and Cost (\$). The 'Revenue (\$)' column is highlighted in green, indicating that a conditional formatting rule has been applied. The rule is set to highlight cells with values greater than 1500. The 'Conditional format rules' panel on the right shows the rule configuration: 'Single color', 'Apply to range' G1:G11, 'Format rules' set to 'Greater than' 1500, and 'Formatting style' set to 'Default' with a green background color.

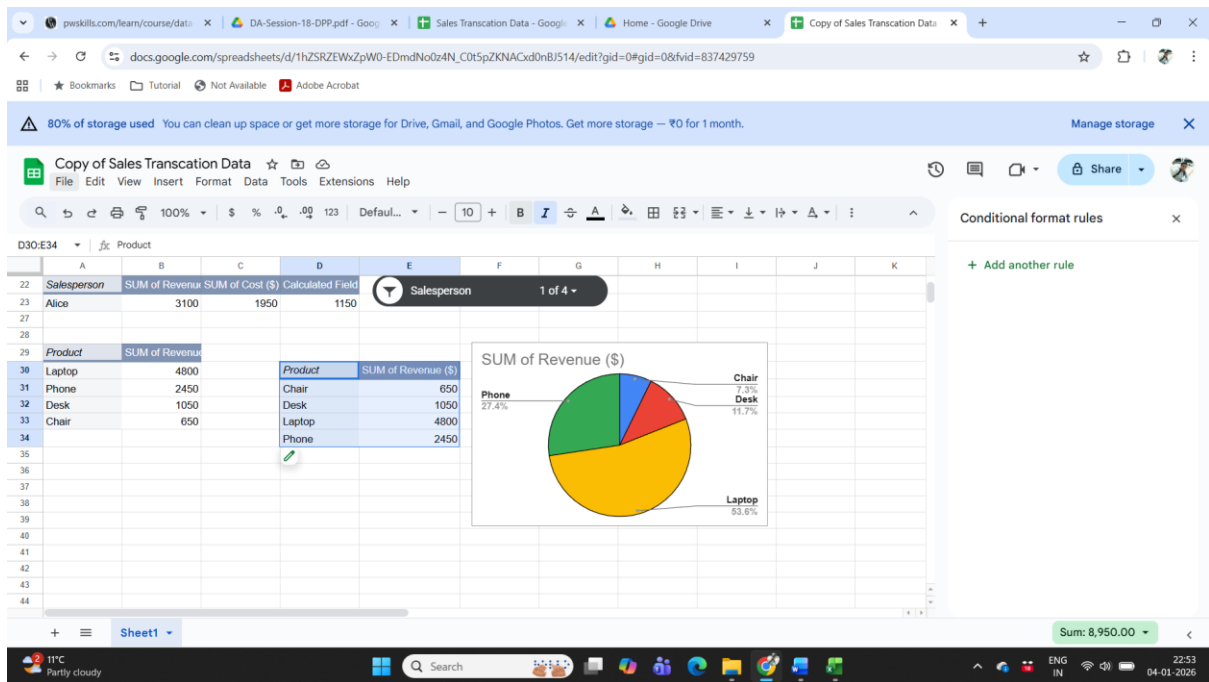
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9. Creating a Pivot Chart

Convert the pivot table into a Pivot Chart to visually compare revenue by product category. Which type of chart would be most suitable?

Ans : Please follow mentioned steps :

1. Select the dataset and click on Insert -> Pivot table option
2. Once the table is inserted, in the row select the Product option
3. In the value field select Revenue summarized by SUM
4. Once the pivot table is created then select the pivot table and select Insert->chart option and select pie chart.



10. Refreshing the Pivot Table

If new transactions are added to the dataset, what steps must you take to update the pivot table so that it reflects the latest data?

Ans : Once any new entry is made into the existing dataset or the dataset is updated then select the pivot table -> make a right click and select the refresh option . Th pivot table would be updated.